TECHNICAL DILIGENCE REVIEW - RECORD 021

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-021

1. Scope, targets and the basis of the capacity assessment

The present report addresses the capacity position, load-test evidence and latency behaviour of the platform at fleet scale. It does not address commercial terms, financial reporting or employment matters, which sit with other workstreams. 2026-06-04 is the evidence cut-off for this report, whose subject is the capacity position, load-test evidence and latency behaviour of the platform at fleet scale. Work on report TDR-2026-25 ran over 4 weeks and included 22 interviews and read-only access to the engineering repositories. Report TDR-2026-25 did not rely on management representation alone; every material claim was traced to a document, a measurement or an observation. The numbering in TDR-2026-25 follows the order of the work rather than the order of importance.

For report TDR-2026-25 the reviewer sampled 3 sites, including the largest deployment and the newest. 21 services were examined in detail for this report, chosen for their position in the dependency map rather than at random. M. Quillan interviewed 22 engineers and managers for TDR-2026-25; two declined to be quoted on one topic and that is noted where it matters. Calder Rowe Engineering Advisory asked the same four questions of every team interviewed for TDR-2026-25, which is how the inconsistencies below were found. Report TDR-2026-25 does not restate the company's own reporting where the reviewer was able to verify the underlying data instead.

M. Quillan tested a sample rather than the population in this report, and the sampling basis is stated in each section. Access for report TDR-2026-25 was sufficient for the conclusions drawn; where it was not, the limitation is stated at the point it bites. This report, this report, is the Calder Rowe Engineering Advisory workstream on the capacity position, load-test evidence and latency behaviour of the platform at fleet scale, delivered to Ardent Peak on 2026-06-04. As instructed, Calder Rowe Engineering Advisory confined report TDR-2026-25 to the capacity position, load-test evidence and latency behaviour of the platform at fleet scale; three adjacent questions raised during the work are listed as follow-ups rather than answered here. The report at TDR-2026-25 was prepared on the footing that a reader may be an engineer or may not, and defines terms where the distinction matters.

M. Quillan treats a backlog ticket as evidence of awareness and a merged change with a test as evidence of remediation, and the report at TDR-2026-25 keeps the two apart. Where report TDR-2026-25 says a defect is verified, the reviewer reproduced it or read a production trace of it; everything else is recorded as a concern. Calder Rowe Engineering Advisory concludes, on the material examined, that the engineering claims that matter to the investment case are supportable, and two of the claims the company leads with are not. M. Quillan's conclusion in this report is that the engineering claims that matter to the investment case are supportable, and two of the claims the company leads with are not. Report TDR-2026-25 is one input to a diligence process and is not a substitute for the company's own engineering judgement.

The present report is limited by log retention: the telemetry window is ninety days, so events before 2025-11-15 could not be examined directly. Two document requests remained partly unsatisfied when the report at TDR-2026-25 was delivered, and the affected conclusions are marked. M. Quillan has left several concerns in TDR-2026-25 at limited evidence strength rather than dropping them or dressing them up. The evidence convention in TDR-2026-25 is that severity describes consequence and evidence strength describes confidence, and the two are never merged. Calder Rowe Engineering Advisory has costed remediation at the company's own loaded engineering rate rather than at a market rate.

22 interviews inform TDR-2026-25. Where an interview is the sole source for a claim, the claim is graded limited. Interviews for report TDR-2026-25 were conducted with the owning engineer present rather than with a manager speaking on their behalf. M. Quillan would add one qualitative item to the pack proposed in report TDR-2026-25: the count of commitments closed without verifying evidence. Monitoring recommended by TDR-2026-25 relies on metrics the company already emits, so no new instrumentation is required to start. The capacity position, load-test evidence and latency behaviour of the platform at fleet scale is a wide field, and this report is explicit about which parts of it were reached.

Schedule J - Site deployments, capacity headroom and integration target (TDR-2026-25)

Site	Customer	Robots deployed	Go-live	Capacity headroom	Warehouse system	Note
Fremont, California	internal engineering site	34	2021-06	not measured	test harness	the reference site, and the only one with a laboratory alongside
Meridian, Idaho	Asterion Logistics	96	2022-09	22%	vendor A	the longest-running customer deployment and the source of the most incident history
Grand Prairie, Texas	Asterion Logistics	78	2023-04	31%	vendor A	the second Asterion site, commissioned on the same template
Wilmington, Ohio	Crestline Fulfillment	112	2023-08	14%	vendor B	the largest deployment and the closest to the allocator scaling limit
Lehigh Valley, Pennsylvania	Crestline Fulfillment	64	2024-02	38%	vendor B	commissioned during the connector rework
Ontario, California	Evergreen Warehousing	88	2024-05	26%	vendor C	the site whose shift pattern change caused the April 2025 charge incident
Groveport, Ohio	Granite Peak Distribution	52	2024-09	44%	vendor A	opportunity charging only, which is where the pack estimator drift appeared
Kenosha, Wisconsin	Dunbar Industrial	41	2024-11	49%	vendor D	the second opportunity-charging site
Tracy, California	Harborline Systems	58	2025-01	36%	vendor B	the site whose cleaning schedule now overlaps the peak shift
Shepherdsville, Kentucky	Fathom Mobility	47	2025-03	41%	vendor C	manipulator-equipped robots are concentrated here
Reno, Nevada	Blue Mesa Components	29	2025-05	57%	vendor D	the smallest customer deployment
Newnan, Georgia	Ironwood Grocery Group	66	2025-07	33%	vendor B	narrow-aisle geometry, and the source of the August 2025 mutual-yield stall
Whitestown, Indiana	Juniper Bay Apparel	44	2025-09	45%	vendor C	the site with the standing map publication workaround
Bethlehem, Pennsylvania	Kestrel Medical Supply	37	2025-11	51%	vendor A	sparse survey features, which is why relocalisation is slow here
Chino, California	Larkspur Home	33	2026-01	54%	vendor D	the go-live delayed by the forked provisioning template
Edwardsville, Illinois	Crestline Fulfillment	24	2026-03	63%	vendor B	the newest Crestline site, still ramping
Lockbourne, Ohio	Granite Peak Distribution	19	2026-04	68%	vendor A	commissioned after the template

Site	Customer	Robots deployed	Go-live	Capacity headroom	Warehouse system	Note
						baseline check landed
Mesquite, Texas	Evergreen Warehousing	16	2026-06	71%	vendor C	in acceptance testing at the review date

2. Load-test programme and the LT-2026-14 result

The percentile set for 720-robot run LT-2026-14 reads not applicable, not applicable and not applicable, with errors at 0.00%. Before reading load run LT-2026-14, note that the environment carries a replayed order stream taken from the two largest sites. Scenario LT-2026-14 is not read as a headline number: the result is consistent with the incident record, which is the strongest corroboration available. scenario LT-2026-14 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Load run LT-2026-15, taken with the other runs in the series, shows that the reviewer would repeat this run with the newest site's order profile before relying on it. The result recorded for scenario LT-2026-15 is pass, which the reviewer traced to the underlying measurement. LT-2026-15 was designed to change one variable, namely the presence of a failure injection. LT-2026-15 is reported at the evidence strength available and has not been promoted to fit the narrative.

Errors in scenario LT-2026-16 were 0.00%, with latency of 1 m 50 s, 4 m 30 s and 9 m 20 s across the three percentiles. Before reading LT-2026-16, note that the environment carries a fleet simulator rather than physical robots, with the control loop unchanged. Scenario LT-2026-16 is recorded as pass with observation. run LT-2026-16 was checked against the corresponding entry in the company's own register, and the register was current.

120-robot run LT-2025-11 is scored against the recovery objective of forty seconds for a database failover, which the company set in its own service documentation. Before reading LT-2025-11, note that the environment carries a smaller order history than production and a synthetic warehouse endpoint. Load run LT-2025-11 exercised the single site, steady state case with 120 robots and 1,850 orders an hour, on 2025-10-08. load run LT-2025-11 is the reason one section of TDR-2026-25 concludes narrowly rather than generally.

Reading 120-robot run LT-2025-12 for the binding constraint gives allocator database write latency became the leading term. Single site, peak hour is the scenario at scenario LT-2025-12: 120 robots, 2,640 orders an hour, dated 2025-10-09. In scenario LT-2025-12 the median was 265 ms and the tail reached 1,240 ms; errors ran at 0.05%. scenario LT-2025-12 is the clearest instance in TDR-2026-25 of engineering practice being better than the documentation suggests.

Measured latency in LT-2025-13 was 310 ms at the median, 870 ms at p95 and 1,690 ms at p99. What 120-robot run LT-2025-13 does and does not establish is worth stating: the reviewer would repeat this run with the newest site's order profile before relying on it. Run LT-2025-13 shows where the system runs out of headroom: charge scheduler retries added measurable queueing. run LT-2025-13 carries a follow-up request that was satisfied after the draft of TDR-2026-25 was circulated.

240-robot run LT-2025-14 is recorded as pass. The test bed for load run LT-2025-14 carries production data volumes and a stubbed paging provider, which is the main reason the reviewer does not treat it as production-equivalent. No limit reached was the constraint 240-robot run LT-2025-14 exposed. 240-robot run LT-2025-14 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Load run LT-2025-15 is the four sites, shared region run, sized at 480 robots and 9,400 orders an hour. Scenario LT-2025-15 exists in the series to isolate the size of the concurrent operator session pool. LT-2025-15 came out pass with observation. 480-robot run LT-2025-15 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Configuration of the environment for load run LT-2025-16 included the current production topology with one region held out. Weighed against the projected fleet, scenario LT-2025-16 indicates that the result holds only for the tested order mix, which is drawn from two sites and is not representative of the newest ones. Load run LT-2025-16 exists in the series to isolate one platform change between otherwise identical runs. 720-robot run LT-2025-16 is included because a reader of TDR-2026-25 should be able to disagree with the inference without repeating the work.

Schedule P - Load-test matrix against the published latency targets (TDR-2026-25)

Run	Scenario	Concurrent robots	Orders per hour	p50	p95	p99	Error rate	Verdict
LT-2025-11	single site, steady state	120	1,850	210 ms	480 ms	760 ms	0.02%	pass
LT-2025-12	single site, peak hour	120	2,640	265 ms	690 ms	1,240 ms	0.05%	pass
LT-2025-13	single site, peak with a charger outage	120	2,640	310 ms	870 ms	1,690 ms	0.11%	pass with observation
LT-2025-14	two sites, shared region	240	4,900	290 ms	820 ms	1,510 ms	0.09%	pass
LT-2025-15	four sites, shared region	480	9,400	355 ms	1,120 ms	2,240 ms	0.18%	pass with observation
LT-2025-16	six sites, shared region	720	13,800	470 ms	1,680 ms	3,410 ms	0.34%	fail against the 1,200 ms p95 target
LT-2025-17	order burst, ten times nominal for two minutes	240	12,000	620 ms	2,350 ms	5,900 ms	1.24%	fail against the 1,200 ms p95 target
LT-2025-18	telemetry burst with control traffic on the shared topic	480	9,400	540 ms	4,100 ms	11,300 ms	0.42%	fail
LT-2025-19	region failover during peak	480	9,400	410 ms	1,950 ms	6,700 ms	2.85%	fail against the error rate target
LT-2025-20	database failover, allocator primary	240	4,900	340 ms	1,410 ms	4,200 ms	0.96%	pass with observation
LT-2026-01	six sites after the topic split	720	13,800	395 ms	1,090 ms	2,050 ms	0.14%	pass
LT-2026-02	eight sites, shared region	960	18,200	455 ms	1,340 ms	2,880 ms	0.27%	fail against the 1,200 ms p95 target
LT-2026-03	eight sites with the allocator read replicas enabled	960	18,200	400 ms	1,150 ms	2,310 ms	0.21%	pass
LT-2026-04	ten sites, shared region	1200	22,600	520 ms	1,720 ms	4,050 ms	0.48%	fail against the 1,200 ms p95 target
LT-2026-05	order burst after the broker queue change	480	24,000	480 ms	1,240 ms	2,760 ms	0.31%	pass with observation
LT-2026-06	shard rebalance under peak load	720	13,800	430 ms	1,280 ms	3,120 ms	0.36%	pass with observation
LT-2026-07	identity service token storm	960	18,200	180 ms	520 ms	1,010 ms	0.03%	pass
LT-2026-08	console session concurrency	480	9,400	290 ms	740 ms	1,480 ms	0.07%	pass
LT-2026-09	console session concurrency at twice the plan	480	9,400	380 ms	1,460 ms	3,900 ms	0.52%	fail
LT-2026-10	warehouse connector burst, one tenant	240	8,800	610 ms	1,830 ms	4,400 ms	0.68%	fail against the 1,500 ms p95 acknowledgement target

Run	Scenario	Concurrent robots	Orders per hour	p50	p95	p99	Error rate	Verdict
LT-2026-11	warehouse connector burst after outbox partitioning	240	8,800	440 ms	1,210 ms	2,540 ms	0.29%	pass
LT-2026-12	telemetry ingest at twice the current fleet	2400	n/a	1.4 s to console	22 s	48 s	0.08%	pass
LT-2026-13	twelve sites, projected 2027 fleet	1440	27,000	690 ms	2,450 ms	6,100 ms	0.74%	fail
LT-2026-14	four hour edge buffer drain after an outage	720	13,800	not applicable	not applicable	not applicable	0.00%	pass with observation
LT-2026-15	map publication to a thousand robots	1000	n/a	3 m 20 s	8 m 40 s	13 m 10 s	0.10%	pass
LT-2026-16	config distribution with thirty per cent of robots offline	720	13,800	1 m 50 s	4 m 30 s	9 m 20 s	0.00%	pass with observation

3. Latency percentiles against the published targets

LT-2025-17 exists in the series to isolate the warehouse system endpoint used. The percentile set for 240-robot run LT-2025-17 reads 620 ms, 2,350 ms and 5,900 ms, with errors at 1.24%. Load run LT-2025-17 exercised the order burst, ten times nominal for two minutes case with 240 robots and 12,000 orders an hour, on 2025-10-21. run LT-2025-17 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Latency in 480-robot run LT-2025-18 was 540 ms at p50, 4,100 ms at p95 and 11,300 ms at p99, with an error rate of 0.42%. The judgement the reviewer draws from telemetry burst with control traffic on the shared topic run LT-2025-18 is that the measurement is credible and the extrapolation the company draws from it is not. Telemetry burst with control traffic on the shared topic is the scenario at scenario LT-2025-18: 480 robots, 9,400 orders an hour, dated 2025-10-22. scenario LT-2025-18 is treated as material to the investment case rather than as a routine engineering item.

480-robot run LT-2025-19 identified the binding constraint: reconnect storm from the edge agents on failover. On region failover during peak run LT-2025-19 the reviewer notes that the result holds only for the tested order mix, which is drawn from two sites and is not representative of the newest ones. LT-2025-19 exists in the series to isolate the presence of a failure injection. 480-robot run LT-2025-19 was inspected in the repository rather than described from the ticket, which is why the account is specific.

Database failover, allocator primary run LT-2025-20, taken with the other runs in the series, shows that the result is consistent with the incident record, which is the strongest corroboration available. Scenario LT-2025-20 was executed in the performance environment, which carries a synthetic robot fleet driven from recorded mission traces. In the series run LT-2025-20 is the run that changes one platform change between otherwise identical runs. scenario LT-2025-20 would be a candidate for the first post-close engineering review under any reasonable plan.

On 720-robot run LT-2026-01 the reviewer notes that the result holds only for the tested order mix, which is drawn from two sites and is not representative of the newest ones. The six sites after the topic split test, recorded as load run LT-2026-01, was executed on 2026-02-11. Scenario LT-2026-01 came out pass. six sites after the topic split run LT-2026-01 was raised again with the owning engineer after the initial interview, and the position did not change.

The reviewer's reading of 960-robot run LT-2026-02 is that the reviewer would repeat this run with the newest site's order profile before relying on it. LT-2026-02 is the eight sites, shared region run, sized at 960 robots and 18,200 orders an hour. Measured latency in scenario LT-2026-02 was 455 ms at the median, 1,340 ms at p95 and 2,880 ms at p99. load run LT-2026-02 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

LT-2026-03 produced a 400 ms median with 1,150 ms and 2,310 ms tails and an 0.21% error rate. Eight sites with the allocator read replicas enabled run LT-2026-03, taken with the other runs in the series, shows that the failure is a capacity finding rather than a defect, and the remediation is a scheduled engineering programme. Eight sites with the allocator read replicas enabled run LT-2026-03 was designed to change one variable, namely whether control and telemetry shared a topic. 960-robot run LT-2026-03 was one of two items where TDR-2026-25 could not obtain the underlying data within the engagement.

LT-2026-04 returned 520 ms median, 1,720 ms at the 95th percentile and 4,050 ms at the 99th, erroring at 0.48%. The production cloud configuration and a laboratory network profile describes the environment LT-2026-04 ran in. Ten sites, shared region run LT-2026-04 is best read against the neighbouring run in the same series, from which it differs in one platform change between otherwise identical runs. 1200-robot run LT-2026-04 would move from a concern to a verified defect if one further measurement were supplied.

Load run LT-2026-05 was marked pass with observation by the performance team. A replayed order stream taken from the two largest sites describes the environment order burst after the broker queue change run LT-2026-05 ran in. In the series 480-robot run LT-2026-05 is the run that changes the warehouse system endpoint used. load run LT-2026-05 was excluded from an earlier draft of TDR-2026-25 and reinstated after the reviewer obtained the supporting record.

Schedule H - Change volume, change failure rate and restore time by month (TDR-2026-25)

Month	Changes to production	Failed changes	Change failure rate	Mean time to restore	Rollbacks
2025-01	41	3	7.3%	34 min	3
2025-02	45	1	2.2%	84 min	2
2025-03	51	3	5.8%	51 min	3
2025-04	59	4	6.7%	50 min	0
2025-05	58	0	0.0%	20 min	0
2025-06	58	2	3.4%	28 min	0
2025-07	42	3	7.1%	35 min	2
2025-08	82	5	6.0%	51 min	1
2025-09	47	0	0.0%	73 min	1
2025-10	62	0	0.0%	32 min	0
2025-11	67	0	0.0%	29 min	1
2025-12	79	1	1.2%	43 min	0
2026-01	70	5	7.1%	24 min	1
2026-02	83	7	8.4%	62 min	1
2026-03	38	1	2.6%	73 min	2
2026-04	44	2	4.5%	59 min	0
2026-05	57	1	1.7%	24 min	1
2026-06	45	4	8.8%	43 min	0
2026-07	64	2	3.1%	74 min	0
2026-08	38	1	2.6%	70 min	0

4. The services that bound throughput

The topology for Site Provisioning Service is cloud, global, and the reviewer confirmed it against the deployment manifests. Service-level intent for Site Provisioning Service states that a new site reaches a working configuration inside one working day. Site Provisioning Service depends on the identity service and the customer network intake records and is depended on by the config distribution service and the WMS connector. Marcus Duplessis leads Platform Infrastructure's work on Site Provisioning Service and was the reviewer's principal interview for it. The open defect backlog for Site Provisioning Service holds 5 items, the most consequential being that site templates diverged after two customers required bespoke network settings. Site Provisioning Service appears twice in the working papers for TDR-2026-25 because the first answer was corrected after a follow-up request.

Go 1.22 is the implementation language for Site Provisioning Service, which is hosted on cloud, containers on a managed Kubernetes cluster with PostgreSQL 14 with a configuration history table behind it. Data ownership is clear for Site Provisioning Service: it writes PostgreSQL 14 with a configuration history table and no other service writes there. Site Provisioning Service was assessed against the objective and the defect record, and no verified defect was identified, and two plausible concerns are recorded below at low evidence strength. Site Provisioning Service is promoted fortnightly, with the ring order fixed by policy. The ageing position for Site Provisioning Service is none identified, and

the reviewer checked it against the vendor statements on file. Site Provisioning Service was cross-checked against the engineering record supplied for TDR-2026-25, and the two agree.

Fleet Allocator depends on the order broker and the robot shadow store and is depended on by the traffic manager, the charge scheduler and the fleet health monitor. 9 items sit in the Fleet Allocator backlog; the entry that bears on the investment case is that assignment starvation for robots parked at chargers below the battery reserve threshold. Fleet Allocator shows 82 per cent line coverage in the pipeline, and the reviewer regards that as adequate for the risk this component carries. Taking the evidence together, Fleet Allocator it is fit for the current fleet size, and the reviewer found no evidence that it would remain so at twice that scale. Hosting for Fleet Allocator is cloud, containers on a managed Kubernetes cluster, and its Go 1.22 code writes through to PostgreSQL 14 with a synchronous standby. Fleet Allocator rests on a single source, and the reviewer has said so rather than implying corroboration.

The topology for Fleet Allocator is cloud, one regional deployment serving all sites in that region, and the reviewer confirmed it against the deployment manifests. The evidence base for Fleet Allocator was the allocator service contract, two design reviews and a week of assignment audit logs. Fleet Allocator releases weekly, behind a change approval. The ageing position for Fleet Allocator is none identified in the reviewed manifests, and the reviewer checked it against the vendor statements on file. Fleet Allocator carries a single measurable objective, namely that task assignment inside 900 ms of order arrival at p95. Fleet Allocator rests on a single source, and the reviewer has said so rather than implying corroboration.

Battery Management Service is owned by the Robot Firmware team, with Peter Halvorsen named as the engineering owner of record. Battery Management Service was examined for unsupported components; one pack revision uses a cell supplier that has since discontinued the part. The published objective for Battery Management Service is that pack cell imbalance is reported before it reaches the derating threshold. Battery Management Service does not share a schema with any consumer; every read by the charge scheduler and the fleet health monitor goes through the published interface. The evidence base for Battery Management Service was the pack firmware release notes, cell test data for two revisions and a derating incident file. Battery Management Service is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Against Battery Management Service the backlog runs to 10 items, and it includes a known behaviour whereby state-of-health estimates drift on packs that spend most of their life on opportunity charge. The topology for Battery Management Service is split edge and cloud, and the reviewer confirmed it against the deployment manifests. Taking the evidence together, Battery Management Service the risk is concentrated in one code path that has neither automated coverage nor a bench equivalent. The Battery Management Service codebase is C on the pack controller, Python 3.11 in the cloud throughout, deployed onto split: pack firmware and a cloud analytics service, and its durable record is pack non-volatile memory and a cloud time-series store. Battery Management Service sits at 54 per cent line coverage; the trend across the reviewed period was up by three points. Battery Management Service was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Config Distribution was built in Rust 1.79 agent with a Go 1.22 control service and has not been rewritten; it still runs on split: an on-robot agent and a cloud service against content-addressed object storage with a PostgreSQL 14 index. Coverage for Config Distribution was 72 per cent line coverage at the review date, and the trend over the reviewed period was flat. Responsibility for Config Distribution was confirmed with Sam Oyelaran, whose team, Platform Infrastructure, also holds the on-call duty. For Config Distribution the stated target is that a configuration change reaches every online robot inside five minutes, expressed against production traffic rather than a benchmark. Config Distribution runs split edge and cloud, with a signed bundle format; that placement is what determines whether a fault is one robot or one region. Config Distribution is the reason one section of TDR-2026-25 concludes narrowly rather than generally.

The site provisioning service and the artefact signing service feed Config Distribution; a failure in Config Distribution is felt by the on-robot supervisor and the motion planner parameters. The evidence base for Config Distribution was the bundle format specification, distribution latency measurements and an offline apply test. 6 items sit in the Config Distribution backlog; the entry that bears on the investment case is that a robot that is offline at publication applies the change without re-validating preconditions. Config Distribution was checked for shared-database coupling, and the ownership boundary held. On end-of-life exposure, Config Distribution shows none; the bundle format was

revised in 2025. Config Distribution is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Safety Controller was checked for shared-database coupling, and the ownership boundary held. Changes to Safety Controller reach production twice a year, with full revalidation of the safety case. Safety Engineering publishes one objective for Safety Controller: a validated protective-stop trigger de-energises the drive inside 120 ms. Safety Controller sits between the E-stop chain, the speed and separation monitor and the drive current sensors and the drive power contactors and the robot fault annunciator, and that position is what makes its availability objective load-bearing. The open defect backlog for Safety Controller holds 3 items, the most consequential being that the fault log wraps after four thousand entries and the oldest records are discarded. Safety Controller is reported at the evidence strength available and has not been promoted to fit the narrative.

The ageing position for Safety Controller is none; the supplier's support commitment runs to 2031, and the reviewer checked it against the vendor statements on file. Underlying Safety Controller in this report are the supplier assessment report, the internal safety case and eight bench trigger tests. Ownership of Safety Controller sits with Safety Engineering; Bettina Vogel signs off changes to it. Safety Controller sits edge, dual-channel, with cross-checking between channels, and no alternative placement is currently supported. Safety Controller is written in C, written to a constrained coding subset, executes on a safety-rated controller module supplied by a third party, and keeps its state in a non-volatile fault log on the controller. Safety Controller is included because a reader of TDR-2026-25 should be able to disagree with the inference without repeating the work.

20 items sit in the Analytics Warehouse Loader backlog; the entry that bears on the investment case is that a late-arriving telemetry partition is silently excluded rather than reprocessed. For Analytics Warehouse Loader the stated target is that the daily load completes before 06:00 local time at every region, expressed against production traffic rather than a benchmark. Analytics Warehouse Loader does not share a schema with any consumer; every read by the customer reporting views and the internal metrics layer goes through the published interface. Farida Cisse leads Data and Telemetry's work on Analytics Warehouse Loader and was the reviewer's principal interview for it. Analytics Warehouse Loader pairs a Python 3.11 with SQL transformations implementation with cloud, a scheduled batch runner, and its data lives in a columnar warehouse with daily partitions. Analytics Warehouse Loader was checked against the corresponding entry in the company's own register, and the register was current.

In production Analytics Warehouse Loader is cloud, one loader per region writing to a shared warehouse, unchanged since the last architecture review. Reading the dependency map from Analytics Warehouse Loader outward gives the telemetry ingest path, the order broker and the billing usage meter on the input side and the customer reporting views and the internal metrics layer on the output side. Documents and data inspected for Analytics Warehouse Loader were the load schedule, partition completeness checks and two months of load audit records. Release cadence for Analytics Warehouse Loader is weekly, and the release records bear that out. Against Analytics Warehouse Loader, the coverage figure is 45 per cent line coverage, which the reviewer regards that as adequate for the risk this component carries. Analytics Warehouse Loader is reported at the evidence strength available and has not been promoted to fit the narrative.

Robot Shadow Store depends on the telemetry ingest path and the edge gateway agent and is depended on by the fleet allocator, the traffic manager and the remote operations console. Ownership of Robot Shadow Store sits with Fleet Services; Marta Kellerman signs off changes to it. 9 defects remain open against Robot Shadow Store; the one the reviewer pressed on is that a shard rebalance briefly serves stale shadow state without marking it stale. End-of-life review of Robot Shadow Store found none identified. The topology for Robot Shadow Store is cloud, regional, sharded by site, and the reviewer confirmed it against the deployment manifests. Robot Shadow Store was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

The objective recorded against Robot Shadow Store is that shadow state is no more than two seconds behind the robot at p95, and it is reported weekly to the engineering leadership. Robot Shadow Store was checked for shared-database coupling, and the ownership boundary held. The Robot Shadow Store codebase is Go 1.22 throughout, deployed onto cloud, containers with a sharded cache, and its durable record is Redis 7 as the hot path with PostgreSQL 14 as the durable record. Robot Shadow Store was assessed on the shadow consistency note, staleness measurements and a rebalance observation, and nothing beyond that was available in the data room. Robot Shadow Store was weighed on its

coverage, its defect backlog and its dependency position, and the component is adequate today and carries a supportability liability that is not on the funded plan. Robot Shadow Store was raised again with the owning engineer after the initial interview, and the position did not change.

Localization Engine shows 66 per cent line coverage in the pipeline, and that is below what this component's failure mode warrants. The Localization Engine codebase is C++17 with a Rust bridge layer throughout, deployed onto on-robot Linux, pinned to a dedicated core, and its durable record is a local map cache with a periodic checkpoint. The interface Localization Engine publishes is versioned, and the motion planner and the fleet health monitor were checked for use of any undocumented field. Release cadence for Localization Engine is every three weeks with the planner, and the release records bear that out. Localization Engine commits that pose error stays under 25 mm in mapped aisles at survey speed, and the objective is measured continuously rather than sampled. Localization Engine was checked against the company's own register, which had not been updated since the change landed.

The ageing position for Localization Engine is an unmaintained scan-matching library is still linked in the fallback path, and the reviewer checked it against the vendor statements on file. Autonomy Core tracks 16 open defects on Localization Engine, including the acknowledged case where relocalisation after a lift-and-place takes longer than the operator prompt allows. Reading the dependency map from Localization Engine outward gives the lidar driver, the wheel odometry filter and the map distribution channel on the input side and the motion planner and the fleet health monitor on the output side. Localization Engine was assessed on the localization design note, survey reports from three sites and the reviewer's map audit, and nothing beyond that was available in the data room. Localization Engine is deployed edge, with map distribution from the cloud. Localization Engine would be a candidate for the first post-close engineering review under any reasonable plan.

Schedule D - Automated coverage trend on the reviewed services (TDR-2026-25)

Service	Coverage 2025-09	Coverage 2026-03	Coverage at review	Change over period	Owning team
Motion Planner	63%	70%	71%	+8	Autonomy Core
Costmap Fusion	51%	55%	58%	+7	Perception
Fleet Allocator	74%	79%	82%	+8	Fleet Services
Traffic Manager	65%	72%	74%	+9	Fleet Services
Charge Scheduler	54%	60%	63%	+9	Fleet Services
Safety Controller	82%	88%	91%	+9	Safety Engineering
E-Stop Chain	n/a	n/a	witnessed test	not comparable	Safety Engineering
Lidar Driver	37%	44%	44%	+7	Robot Firmware
Localization Engine	61%	66%	66%	+5	Autonomy Core
WMS Connector	60%	68%	69%	+9	Integrations
Order Broker	72%	75%	78%	+6	Fleet Services
Telemetry Ingest	55%	58%	61%	+6	Data and Telemetry
OTA Updater	67%	73%	76%	+9	Platform Infrastructure
Identity and Authorization	77%	82%	84%	+7	Security Engineering
Remote Operations Console	48%	51%	52%	+4	Applications
Simulation Harness	44%	46%	47%	+3	Developer Productivity
Hardware-in-the-Loop Rig	35%	36%	39%	+4	Robot Firmware

4. The services that bound throughput - part 2 of 2

This is part 2 of the 4. The services that bound throughput, carried over unbroken; a reader reconciling a reference should treat the parts as a single register.

No basis of compilation changes at this break; whatever the 4. The services that bound throughput recorded about scope and method at its head continues to apply. Nothing is restated from the earlier part.

Schedule D - Automated coverage trend on the reviewed services (TDR-2026-25) (continued, part 2 of 2)

Service	Coverage 2025-09	Coverage 2026-03	Coverage at review	Change over period	Owning team
Fleet Health Monitor	50%	54%	57%	+7	Field Reliability
Battery Management Service	45%	51%	54%	+9	Robot Firmware
Gripper Controller	38%	38%	41%	+3	Robot Firmware

Service	Coverage 2025-09	Coverage 2026-03	Coverage at review	Change over period	Owning team
Map Authoring Studio	43%	48%	49%	+6	Applications
Site Provisioning Service	55%	62%	64%	+9	Platform Infrastructure
Event Bus	n/a	n/a	witnessed test	not comparable	Platform Infrastructure
Config Distribution	68%	69%	72%	+4	Platform Infrastructure
Robot Shadow Store	63%	68%	68%	+5	Fleet Services
Analytics Warehouse Loader	42%	43%	45%	+3	Data and Telemetry
Billing Usage Meter	75%	77%	79%	+4	Data and Telemetry
Alarm Router	46%	50%	51%	+5	Field Reliability
Video Snippet Service	37%	43%	43%	+6	Applications
Speed and Separation Monitor	82%	87%	87%	+5	Safety Engineering
Wheel Odometry Filter	53%	60%	62%	+9	Robot Firmware
Payload Scale Interface	29%	33%	36%	+7	Robot Firmware
Edge Gateway Agent	67%	73%	74%	+7	Platform Infrastructure
Release Orchestrator	47%	53%	55%	+8	Developer Productivity

5. Capacity headroom site by site

Capacity headroom at Bethlehem, Pennsylvania is 51%, and the warehouse system is vendor A. Bethlehem, Pennsylvania is sparse survey features, which is why relocalisation is slow here, and the reviewer weighted the sample accordingly. Bethlehem, Pennsylvania was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

88 robots are deployed at Ontario, California, which went live in 2024-05. Against its peak profile, Ontario, California retains 26% headroom on a vendor C integration. Ontario, California is the entry that changed most between the first and second document requests behind TDR-2026-25.

What distinguishes Meridian, Idaho is that it is the longest-running customer deployment and the source of the most incident history. 2022-09 is the go-live month at Meridian, Idaho, where Asterion Logistics now runs 96 robots. Meridian, Idaho was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

For Fremont, California the material fact is that it is the reference site, and the only one with a laboratory alongside. Fremont, California runs 34 robots for internal engineering site and opened in 2021-06. Fremont, California is the entry that changed most between the first and second document requests behind TDR-2026-25.

Asterion Logistics operates 78 Helioscope robots at Grand Prairie, Texas, commissioned in 2023-04. Grand Prairie, Texas is notable in this review because it is the second Asterion site, commissioned on the same template. Grand Prairie, Texas was checked against the company's own register, which had not been updated since the change landed.

Capacity headroom at Edwardsville, Illinois is 63%, and the warehouse system is vendor B. Site ST-16 is Edwardsville, Illinois: Crestline Fulfillment, 24 robots, live from 2026-03. Edwardsville, Illinois is reported at the evidence strength available and has not been promoted to fit the narrative.

2024-02 is the go-live month at Lehigh Valley, Pennsylvania, where Crestline Fulfillment now runs 64 robots. Lehigh Valley, Pennsylvania is commissioned during the connector rework, and the reviewer weighted the sample accordingly. Lehigh Valley, Pennsylvania appears twice in the working papers for TDR-2026-25 because the first answer was corrected after a follow-up request.

Chino, California is notable in this review because it is the go-live delayed by the forked provisioning template. Chino, California came online in 2026-01 and now carries 33 robots for Larkspur Home. Chino, California was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Kenosha, Wisconsin came online in 2024-11 and now carries 41 robots for Dunbar Industrial. For Kenosha, Wisconsin the material fact is that it is the second opportunity-charging site. Kenosha, Wisconsin was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

What distinguishes Whitestown, Indiana is that it is the site with the standing map publication workaround. The Whitestown, Indiana deployment serves Juniper Bay Apparel with 44 robots, live since 2025-09. Whitestown, Indiana is the clearest instance in TDR-2026-25 of engineering practice being better than the documentation suggests.

Remaining capacity at Reno, Nevada is 57%, measured against the site's peak order profile. At Reno, Nevada the fleet is 29 robots, and the site has been live since 2025-05. Reno, Nevada was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

The Groveport, Ohio site integrates with vendor A and shows 44% headroom. On Groveport, Ohio the record shows that it is opportunity charging only, which is where the pack estimator drift appeared. Groveport, Ohio has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

For Tracy, California the material fact is that it is the site whose cleaning schedule now overlaps the peak shift. Tracy, California reports 36% headroom against its integration with vendor B. Tracy, California is reported here in full because a summary of it in an earlier draft of TDR-2026-25 was misread as an assurance.

Newnan, Georgia is narrow-aisle geometry, and the source of the August 2025 mutual-yield stall, and the reviewer weighted the sample accordingly. Site ST-12 is Newnan, Georgia: Ironwood Grocery Group, 66 robots, live from 2025-07. Newnan, Georgia is included because a reader of TDR-2026-25 should be able to disagree with the inference without repeating the work.

Mesquite, Texas is in acceptance testing at the review date, and the reviewer weighted the sample accordingly. Mesquite, Texas reports 71% headroom against its integration with vendor C. Mesquite, Texas is treated as material to the investment case rather than as a routine engineering item.

Wilmington, Ohio runs 112 robots for Crestline Fulfillment and opened in 2023-08. 14% of headroom remains at Wilmington, Ohio, whose connector targets vendor B. Wilmington, Ohio would move from a concern to a verified defect if one further measurement were supplied.

For Lockbourne, Ohio the material fact is that it is commissioned after the template baseline check landed. Vendor A is the warehouse system at Lockbourne, Ohio, where headroom is 68%. Lockbourne, Ohio has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

6. Capacity incidents and what they revealed

Dunbar Industrial felt INC-2026-057, which affected 2 sites and 5 robots. On delivery, modelling event INC-2026-057 shows this: not landed; data collection is running and the imbalance alert is in review. INC-2026-057 would have been shorter or smaller had it not been that the two affected sites moved to opportunity charging after the estimator had been validated. On Dunbar Industrial incident INC-2026-057 the evidence is strong, resting on the production traces the team supplied; the severity assessment is stated separately below. Battery Management Service incident INC-2026-057 was detected because a robot derated mid-shift and the pack was pulled for analysis. Dunbar Industrial incident INC-2026-057 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Modelling incident INC-2026-057 supports one conclusion and not another: management's severity rating is defensible and its remediation priority is not. The engineering team's root cause for modelling incident INC-2026-057 is that the estimator was fitted on packs that follow a full charge cycle, and the reviewer found the analysis sound. Modelling event INC-2026-057 shows a short mitigation and a long tail: not a service outage; robots derated against open at the review date. Dunbar Industrial incident INC-2026-057 records that on 2026-05-19, state-of-health estimates on opportunity-charged packs drifted enough to hide a cell imbalance until derating. Battery Management Service accounts for modelling event INC-2026-057, a modelling event. Dunbar Industrial incident INC-2026-057 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

INC-2025-058 produced one commitment: to key suppression on the tenant and identifier pair and backfill the affected order records, owned by Tomas Villareal, dated 2025-07-15. Moderate, resting on the team's own post-mortem and one supporting log extract is the evidence grade for data integrity event INC-2025-058, and the reader should not read

severity into it. For incident INC-2025-058 the follow-through was tested rather than assumed, and landed on 2025-05-30; the backfill was completed on 2025-06-03. WMS Connector incident INC-2025-058 took 2 h 35 m to stabilise and 21 days to close out. INC-2025-058 belongs to the data integrity group and sits with WMS Connector. data integrity event INC-2025-058 illustrates why TDR-2026-25 separates the severity of a finding from the confidence behind it.

Root cause in Dunbar Industrial incident INC-2025-058 was that duplicate suppression keyed on a customer-supplied identifier alone rather than on tenant and identifier. In data integrity incident INC-2025-058 the first signal was the customer reported missing picks the same afternoon, and the reviewer treats that as reported detection. On 2025-05-09 the record for INC-2025-058 opens: two tenants reused the same order identifier namespace and duplicate suppression discarded live orders. Dunbar Industrial incident INC-2025-058 affected 2 site or tenant with no robot taken out of service. In Dunbar Industrial incident INC-2025-058, the integration specification allowed the identifier to be tenant-scoped but did not require it, which is a process finding rather than a code finding. WMS Connector incident INC-2025-058 was raised again with the owning engineer after the initial interview, and the position did not change.

The classification applied to Charge Scheduler incident INC-2025-052 is scheduling, and the affected component is Charge Scheduler. Evergreen Warehousing incident INC-2025-052 is recorded as follows on delivery: landed on 2025-06-11 with a nightly reconciliation report. Scheduling event INC-2025-052 affected one site and 44 robots. Incident INC-2025-052 was caused by a design gap: the scheduler treated a shift calendar entry as immutable once the window had opened. Evergreen Warehousing incident INC-2025-052 was raised on 2025-04-21 after an early shift end left opportunity-charge windows reserved and the following shift started with a depleted fleet. scheduling event INC-2025-052 is the reason one section of TDR-2026-25 concludes narrowly rather than generally.

Grading scheduling incident INC-2025-052 on evidence rather than on consequence gives limited; the log retention window had closed before the reviewer asked for the raw data. The reviewer treats one contributing factor in scheduling event INC-2025-052 as the more important lesson: the site had changed its shift pattern without notifying the deployment team. Scheduling incident INC-2025-052 was detected because the site's own shift report, escalated the next morning. Following incident INC-2025-052 the team undertook to release charge windows on a shift-end event and reconcile the calendar nightly, with Nadia Semenova accountable by 2025-06-30. What INC-2025-052 tells the reviewer is that the failure was in the detection design rather than in the code that failed. scheduling incident INC-2025-052 was excluded from an earlier draft of TDR-2026-25 and reinstated after the reviewer obtained the supporting record.

Classifying Kestrel Medical Supply incident INC-2025-110 as data integrity matters because the Analytics Warehouse Loader backlog contains two further items of the same kind. Attention came to data integrity incident INC-2025-110 when the customer challenged the report three days later. Analytics Warehouse Loader incident INC-2025-110 touched 1 site or tenant with no robot taken out of service, and Kestrel Medical Supply was the affected account. INC-2025-110 carries limited; the log retention window had closed before the reviewer asked for the raw data evidence, and the reviewer declined to raise the finding beyond what that supports. INC-2025-110 closure status is landed on 2025-09-21 with a reprocessing run for the affected week. data integrity event INC-2025-110 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

The corrective action for Analytics Warehouse Loader incident INC-2025-110 is to fail the load on an incomplete partition and reprocess the affected days, and 2025-10-31 is the date the team gave. The judgement recorded against Analytics Warehouse Loader incident INC-2025-110 is that the failure was in the detection design rather than in the code that failed. Kestrel Medical Supply incident INC-2025-110 was caused by a design gap: the loader treated a missing partition as empty rather than as incomplete. Contributing to incident INC-2025-110, and not repaired by the remediation, is that the partition completeness check was written but never wired into the load gate. Reading Analytics Warehouse Loader incident INC-2025-110 from its first entry, on 2025-09-03 a late telemetry partition was excluded from the daily load and a customer report understated throughput. Kestrel Medical Supply incident INC-2025-110 was checked against the company's own register, which had not been updated since the change landed.

The immediate event in incident INC-2025-071 was that a regional ingest endpoint failed over and edge buffers filled during a fifty-two minute outage, dated 2025-06-11. The affected population in availability incident INC-2025-071 was 6 sites and 214 robots. The reviewer grades the Telemetry Ingest incident INC-2025-071 evidence as strong on the cause and limited on the contributing factor and has not upgraded the finding to compensate. What went wrong in Telemetry Ingest incident INC-2025-071 was that a broker partition lost its leader during a rolling upgrade of the

streaming cluster. Assessed on its own record, multiple sites incident INC-2025-071 shows that the failure was in the detection design rather than in the code that failed. incident INC-2025-071 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

52 m elapsed before Telemetry Ingest incident INC-2025-071 was mitigated, with resolution at 52 m. Telemetry Ingest owns availability incident INC-2025-071, which the reviewer classes as availability. Erik Palmgren committed after availability incident INC-2025-071 to add a leadership and under-replicated partition check to the broker upgrade runbook, due 2025-07-31. How availability event INC-2025-071 surfaced matters to the assessment: the ingest availability alert fired within ninety seconds. The contributing factor recorded against availability incident INC-2025-071 is that the upgrade runbook did not require a partition leadership check between broker restarts. incident INC-2025-071 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Robot Shadow Store accounts for Robot Shadow Store incident INC-2025-124, a consistency event. In consistency incident INC-2025-124, no test exercised a shard rebalance while the cluster was under load, which is a process finding rather than a code finding. Time to mitigate INC-2025-124 was 1 h 08 m, and time to resolve was 29 days. Consistency event INC-2025-124 rests on limited; the log retention window had closed before the reviewer asked for the raw data evidence, which the reviewer states separately from the severity of the finding. On delivery, consistency event INC-2025-124 shows this: landed on 2025-10-30; the load test was added on 2025-11-12. Robot Shadow Store incident INC-2025-124 was raised again with the owning engineer after the initial interview, and the position did not change.

Attention came to Robot Shadow Store incident INC-2025-124 when the console latency alert fired, though the underlying cause took a day to attribute. The corrective action for incident INC-2025-124 is to annotate staleness on every read path and add a rebalance-under-load test to the platform suite, and 2025-11-30 is the date the team gave. Drawing the evidence in incident INC-2025-124 together, the same class of defect can recur in two sibling components that were not changed. The affected population in consistency incident INC-2025-124 was 5 sites and 178 robots. What went wrong in INC-2025-124 was that the rebalance path bypassed the staleness annotation applied on the normal read path. consistency incident INC-2025-124 is the entry that changed most between the first and second document requests behind TDR-2026-25.

The reviewer's reading of INC-2026-028 is that the same class of defect can recur in two sibling components that were not changed. INC-2026-028 would have been shorter or smaller had it not been that case closure did not check for unresolved constituent alerts. Time to mitigate observability incident INC-2026-028 was not a service outage; 2 days of delayed repair, and time to resolve was 31 days. Reported detection applies to observability event INC-2026-028, since the customer asked why one symptom persisted after the case had been closed. Grading Granite Peak Distribution incident INC-2026-028 on evidence rather than on consequence gives moderate on the cause and strong on the delivery position, which was checked in the repository. INC-2026-028 was inspected in the repository rather than described from the ticket, which is why the account is specific.

Reading Fleet Health Monitor incident INC-2026-028 from its first entry, on 2026-03-10 two distinct faults at one site were grouped into a single case and the second went unworked for two days. The remediation committed after observability incident INC-2026-028 was to group on the signal and block closure while a constituent alert remains active, owned by Wanda Njoku with a due date of 2026-04-30. Classifying observability event INC-2026-028 as observability matters because the Fleet Health Monitor backlog contains two further items of the same kind. So far as reach is concerned, observability incident INC-2026-028 involved one site and 22 robots. On delivery, observability incident INC-2026-028 shows this: landed on 2026-04-10. Fleet Health Monitor incident INC-2026-028 was checked against the corresponding entry in the company's own register, and the register was current.

In observability incident INC-2026-045 the first signal was the buffer pressure metric was visible per robot but no site-level alert existed, and the reviewer treats that as alerted detection. The post-mortem for incident INC-2026-045 identifies the cause as follows: the buffer is sized for four hours and drops oldest-first without raising an alert. Observability is the class of incident INC-2026-045; Edge Gateway Agent is the component. In Edge Gateway Agent incident INC-2026-045, buffer pressure was never aggregated to a site view, so the condition was not observable, which is a process finding rather than a code finding. The commitment register shows Evergreen Warehousing incident INC-2026-045 against an undertaking to aggregate buffer pressure to a site alert and extend the buffer to twelve hours on

the current hardware, with Sam Oyelaran named and 2026-06-30 committed. incident INC-2026-045 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Mitigation of observability event INC-2026-045 took 5 h 12 m and full resolution took 26 days. Verification of the INC-2026-045 remediation gives the alert landed on 2026-05-17; the buffer extension is in progress. Edge Gateway Agent incident INC-2026-045 records that on 2026-04-21, a site-wide network change left robots buffering for five hours and the oldest telemetry samples were discarded. The blast radius of observability incident INC-2026-045 was one site and 39 robots. The judgement recorded against incident INC-2026-045 is that the blast radius was set by a deployment topology decision made years earlier. Edge Gateway Agent incident INC-2026-045 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Schedule F - Incident population examined, with follow-through status (TDR-2026-25)

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
INC-2025-041	2025-03-14	Traffic Manager	2	61	1 h 52 m	9 days	resource leak	landed
INC-2025-047	2025-04-02	Lidar Driver	1	12	38 m	3 days	timing	partly landed
INC-2025-052	2025-04-21	Charge Scheduler	1	44	4 h 10 m	16 days	scheduling	landed
INC-2025-058	2025-05-09	WMS Connector	2	0	2 h 35 m	21 days	data integrity	landed
INC-2025-063	2025-05-27	Fleet Allocator	3	28	6 h 05 m	12 days	policy defect	landed
INC-2025-071	2025-06-11	Telemetry Ingest	6	214	52 m	52 m	availability	landed
INC-2025-078	2025-06-24	Remote Operations Console	4	0	1 h 15 m	34 days	stale state	landed
INC-2025-084	2025-07-08	OTA Updater	1	11	27 m	5 days	state persistence	landed
INC-2025-090	2025-07-19	Safety Controller	1	3	immediate stop, 22 m to return to service	41 days	hardware fault	partly landed
INC-2025-096	2025-08-05	Alarm Router	0	0	not applicable; no service impact	26 days	process gap	partly landed
INC-2025-103	2025-08-22	Motion Planner	1	2	19 m	58 days	algorithm defect	partly landed
INC-2025-110	2025-09-03	Analytics Warehouse Loader	1	0	not applicable; reporting only	18 days	data integrity	landed
INC-2025-118	2025-09-17	Identity and Authorizatio n	0	0	3 h 40 m to revoke and rotate	7 days	secret exposure	partly landed
INC-2025-124	2025-10-01	Robot Shadow Store	5	178	1 h 08 m	29 days	consistency	landed
INC-2025-131	2025-10-14	Map Authoring Studio	1	34	3 h 22 m	44 days	process bypass	partly landed

6. Capacity incidents and what they revealed - part 2 of 2

Following on directly, these further rows of the 6. Capacity incidents and what they revealed belong to the same schedule and share its numbering. Pagination alone determines where it falls.

Readers should carry the 6. Capacity incidents and what they revealed's stated conventions across the break, since this part adds no definition of its own. Pagination alone determines where it falls.

Schedule F - Incident population examined, with follow-through status (TDR-2026-25) (continued, part 2 of 2)

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
INC-2025-	2025-11-02	Event Bus	7	246	2 h 47 m	63 days	capacity	partly landed

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
139								
INC-2025-146	2025-11-19	Billing Usage Meter	0	0	not applicable; caught before billing	11 days	process defect	landed
INC-2025-152	2025-12-04	Video Snippet Service	0	0	not applicable; retention only	37 days	data retention	partly landed
INC-2026-004	2026-01-13	Localization Engine	1	18	not a service outage; 5 days to a workaround	open at the review date	performance	not landed
INC-2026-009	2026-01-27	Gripper Controller	1	4	1 h 30 m to quarantine the affected robots	48 days	calibration	landed
INC-2026-015	2026-02-09	Config Distribution	2	7	44 m	22 days	validation	landed
INC-2026-021	2026-02-24	Simulation Harness	0	0	not applicable; developer tooling	open at the review date	tooling	not landed
INC-2026-028	2026-03-10	Fleet Health Monitor	1	22	not a service outage; 2 days of delayed repair	31 days	observability	landed
INC-2026-033	2026-03-25	Speed and Separation Monitor	1	6	immediate speed reduction; 3 h to review the fleet	open at the review date	safety margin	not landed
INC-2026-039	2026-04-07	Order Broker	1	1	1 h 04 m	19 days	concurrency	landed
INC-2026-045	2026-04-21	Edge Gateway Agent	1	39	5 h 12 m	26 days	observability	partly landed
INC-2026-051	2026-05-06	Site Provisioning Service	1	0	2 h 18 m	9 days	configuration drift	partly landed
INC-2026-057	2026-05-19	Battery Management Service	2	5	not a service outage; robots derated	open at the review date	modelling	not landed
INC-2026-062	2026-06-02	Release Orchestrator	0	0	not applicable; process only	open at the review date	process control	not landed
INC-2026-068	2026-06-16	Wheel Odometry Filter	1	14	not a service outage; 3 h of degraded operation	open at the review date	environmental	not landed

7. The scalability programme, its sequence and its cost

Sizing on Simulation Harness item TD-06 gives 1 engineer-quarter and $115,000. The reviewer's qualification on programme item TD-06 is that blocked on the physics core determinism work. Reading the programme plan, programme item TD-06 is behind a supplier release outside the company's control. register item TD-06 would move from a concern to a verified defect if one further measurement were supplied.

TD-03 on Costmap Fusion cannot start earlier than it does because it is behind a platform change that has its own target date. Reading programme item TD-03 against the funded plan, management ranked this behind the separation monitoring work. The register entry Costmap Fusion item TD-03 reads: retire the vendored geometry library in costmap fusion, in the support category. support item TD-03 was one of two items where TDR-2026-25 could not obtain the underlying data within the engagement.

The dependency structure places Simulation Harness item TD-07 last in a three-item chain, so its date is the least reliable on the register. Left in place, the condition behind register item TD-07 non-reproducible test results undermine every simulation claim. Register item TD-07 is the item to pin the full simulation build configuration and enable a determinism check, classified under tooling. tooling item TD-07 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

TD-12 on Edge Gateway Agent is in progress with a target date of 2026-10-31. Funding for TD-12 on Edge Gateway Agent is unfunded and dependent on capital rather than engineering effort, and the register does not say so. Reading resilience item TD-12 against the funded plan, the site alert landed first; the buffer extension needs a storage change on the compute module. register item TD-12 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The dependency structure places TD-16 on Localization Engine after the determinism work in the simulation harness. 1 engineer-quarter is the estimate on programme item TD-16, which converts to $126,000. The register entry programme item TD-16 reads: encrypt the on-robot localization map cache, in the security category. Localization Engine item TD-16 is included because a reader of TDR-2026-25 should be able to disagree with the inference without repeating the work.

Exposure carried by register item TD-02 is that it strong-copyleft exposure in shipped robot software plus an unsupported dependency. On funding, programme item TD-02 is funded by reallocating effort from feature work. Sizing on Localization Engine item TD-02 gives 1.5 engineer-quarters and $185,000. register item TD-02 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Raised in the 2026 assessment and accepted for remediation is the point the reviewer would put to management on TD-15. Reading the programme plan, Alarm Router item TD-15 is last in a three-item chain, so its date is the least reliable on the register. Add per-region credentials for the paging provider is entered at TD-15 on Alarm Router as an security item. programme item TD-15 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Reading the programme plan, programme item TD-09 is in parallel with the release-process changes. Programme item TD-09 is on the register because it an unsupported broker underneath every service. About Event Bus item TD-09 it should be said that the upgrade requires a maintenance window at every site. programme item TD-09 is the entry that changed most between the first and second document requests behind TDR-2026-25.

The register shows TD-13 as in design, due 2026-12-31. The consequence of leaving register item TD-13 undone is that it relocalisation exceeds the operator prompt timeout at sparse sites. Reading the programme plan, TD-13 on Localization Engine is behind a customer maintenance window that is not yet agreed. TD-13 illustrates why TDR-2026-25 separates the severity of a finding from the confidence behind it.

0.5 engineer-quarters is the estimate on TD-05, which converts to $96,000. Funding for TD-05 on Hardware-in-the-Loop Rig is partly funded, with the second phase unbudgeted, and the register does not say so. Committed is the status of TD-05, targeted at 2026-09-30. register item TD-05 is treated as material to the investment case rather than as a routine engineering item.

The plan treats register item TD-04 as not funded, and carried in the register regardless, which the reviewer checked against the headcount model. Behind a customer maintenance window that is not yet agreed is where quality item TD-04 sits in the programme. As at the review date Costmap Fusion item TD-04 was in progress, with 2027-03-31 still the stated target. register item TD-04 is one of the entries where TDR-2026-25 relies on the company's own measurement rather than an independent one.

Effort on Charge Scheduler item TD-23 is estimated at 0.75 engineer-quarters, costed at $92,000. Split the shared database instance used by the allocator and the charge scheduler is entered at reliability item TD-23 as an reliability item. Behind a supplier release outside the company's control is where register item TD-23 sits in the programme. TD-23 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Analytics Warehouse Loader item TD-10 carries an estimate of 0.5 engineer-quarters; the costed figure is $54,000. The reviewer's comment on support item TD-10 is that the warehouse vendor has published a withdrawal date. The plan treats TD-10 as unfunded and dependent on capital rather than engineering effort, which the reviewer checked against the

headcount model. TD-10 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Under quality, entry quality item TD-21 is to refit the battery state-of-health estimator on opportunity-charge data. Quality item TD-21 is not straightforward: data collection is running at the two affected sites. Partly funded, with the second phase unbudgeted is the funding position on quality item TD-21. Battery Management Service item TD-21 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

What programme item TD-19 protects against is the fact that it the protective distance rests on a fixed approach speed assumption. The register entry register item TD-19 reads: measure closing rate for the separation monitor, in the safety category. On programme item TD-19 the reviewer records that the most significant open safety item in the reviewed record. safety item TD-19 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Programme item TD-20 is not straightforward: the prototype uses a network-copyleft component that must be replaced before shipping. Left in place, the condition behind TD-20 on Wheel Odometry Filter drift on wet floors is neither corrected nor flagged. The plan treats TD-20 as funded in the current financial year, which the reviewer checked against the headcount model. TD-20 on Wheel Odometry Filter is one of the entries where TDR-2026-25 relies on the company's own measurement rather than an independent one.

Programme item TD-18 covers the work to aggregate the safety controller fault log to the cloud continuously. Left in place, the condition behind programme item TD-18 investigation evidence is lost between weekly extractions. The reviewer's comment on TD-18 on Safety Controller is that the safety team supports it; it is not funded in the current plan. Safety Controller item TD-18 appears twice in the working papers for TDR-2026-25 because the first answer was corrected after a follow-up request.

The security supportability entry at security supportability item TD-08 is to replace the archived charting library in the operator console. As at the review date TD-08 was backlog, with 2027-01-31 still the stated target. Security supportability item TD-08 is sequenced last in a three-item chain, so its date is the least reliable on the register. register item TD-08 is one of the entries where TDR-2026-25 relies on the company's own measurement rather than an independent one.

8. Resilience runs: failover, rebalance and buffer drain

The environment behind LT-2026-06 carries a fleet simulator rather than physical robots, with the control loop unchanged, and the reviewer weighted the result accordingly. Scenario LT-2026-06 supports this conclusion: the result is consistent with the incident record, which is the strongest corroboration available. The constraint surfaced by shard rebalance under peak load run LT-2026-06 is the staleness annotation added after the October 2025 incident held. scenario LT-2026-06 illustrates why TDR-2026-25 separates the severity of a finding from the confidence behind it.

Scenario LT-2026-07 was judged against the recovery objective of forty seconds for a database failover. Weighed against the projected fleet, 960-robot run LT-2026-07 indicates that the run was executed against a synthetic warehouse system rather than a customer endpoint, which flatters the tail. The verdict on LT-2026-07 is pass. LT-2026-07 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Against the stated targets, run LT-2026-08 is pass. Run LT-2026-08 exists in the series to isolate the size of the concurrent operator session pool. Run LT-2026-08 is the console session concurrency run, sized at 480 robots and 9,400 orders an hour. scenario LT-2026-08 is the entry that changed most between the first and second document requests behind TDR-2026-25.

1,460 ms at p95 is the figure that matters in scenario LT-2026-09, against 380 ms median and 3,900 ms at the tail. Measurement of run LT-2026-09 is against the internal objective that no telemetry is lost for an outage under four hours rather than against a customer commitment. Fail is what the test record gives for run LT-2026-09. scenario LT-2026-09 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

The point of LT-2026-10 within the series is whether control and telemetry shared a topic. Warehouse connector burst, one tenant run LT-2026-10, taken with the other runs in the series, shows that the result is consistent with the incident record, which is the strongest corroboration available. Errors in load run LT-2026-10 were 0.68%, with latency

of 610 ms, 1,830 ms and 4,400 ms across the three percentiles. load run LT-2026-10 carries a follow-up request that was satisfied after the draft of TDR-2026-25 was circulated.

The test bed for LT-2026-11 carries production-sized data but a single availability zone, which is the main reason the reviewer does not treat it as production-equivalent. Load run LT-2026-11 returned 440 ms median, 1,210 ms at the 95th percentile and 2,540 ms at the 99th, erroring at 0.29%. Pass is what the test record gives for LT-2026-11. load run LT-2026-11 is included because a reader of TDR-2026-25 should be able to disagree with the inference without repeating the work.

Run LT-2026-12 used the standard performance environment, which is characterised by a fleet simulator rather than physical robots, with the control loop unchanged. Scenario LT-2026-12 identified the binding constraint: the ingest path scales horizontally with no shared bottleneck. In the series run LT-2026-12 is the run that changes whether control and telemetry shared a topic. run LT-2026-12 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

1440 robots and 27,000 orders an hour define 1440-robot run LT-2026-13, run on 2026-04-14. Whether the read replicas were enabled is what separates twelve sites, projected 2027 fleet run LT-2026-13 from the adjacent run. On run LT-2026-13 the reviewer notes that the reviewer would repeat this run with the newest site's order profile before relying on it. run LT-2026-13 was checked against the corresponding entry in the company's own register, and the register was current.

9. Dependencies with a capacity implication

In Event Bus the component grpc-transport is used service-to-service transport across the cloud, which is the fact the licence analysis rests on. Against grpc-transport the reviewer would add that no notice obligations beyond attribution. Version 1.62.0 of grpc-transport is the catalogued build at OSS-004, used service-to-service transport across the cloud. grpc-transport was one of two items where TDR-2026-25 could not obtain the underlying data within the engagement.

Yaml-parse carries this qualification: attribution present. No copyleft applies to yaml-parse, whose licence is MIT. Compliance with the yaml-parse terms requires nothing beyond retaining the licence text with the source. yaml-parse is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Where k8s-operator-framework came from matters here: public registry, signature verified. The non-licence risk on k8s-operator-framework is a fork that must be maintained locally for as long as it is used. On k8s-operator-framework the reviewer records that attribution present. k8s-operator-framework is reported here in full because a summary of it in an earlier draft of TDR-2026-25 was misread as an assurance.

Licensing for otel-instrument is Apache-2.0 and the copyleft column reads no. The operational obligation attaching to otel-instrument is to hold a current commercial licence for each developer seat. Version 1.24.0 of otel-instrument is the catalogued build at OSS-023, used tracing across cloud services. otel-instrument was excluded from an earlier draft of TDR-2026-25 and reinstated after the reviewer obtained the supporting record.

Wms-adapter-sdk entered the codebase by this route: supplied by the warehouse system vendor under a written licence. Wms-adapter-sdk carries commercial, per-seat terms, classified no for copyleft purposes. Wms-adapter-sdk shows an approval status of approved 2023-12-05 against the company's own policy. wms-adapter-sdk was checked against the corresponding entry in the company's own register, and the register was current.

Geom2d-fusion entered the codebase by this route: public registry; upstream branch unmaintained since 2024. Geom2d-fusion affects Costmap Fusion; no other component reviewed depends on it. Meeting the terms on geom2d-fusion means to share alike any derived asset it distributes. geom2d-fusion illustrates why TDR-2026-25 separates the severity of a finding from the confidence behind it.

The licence on rtos-shim is BSD-2-Clause; copyleft exposure is no. In Motion Planner the component rtos-shim is used real-time helper layer on the robot, which is the fact the licence analysis rests on. Rtos-shim carries a practical obligation: to attribute the component in the customer-facing notice file. rtos-shim appears twice in the working papers for TDR-2026-25 because the first answer was corrected after a follow-up request.

Provenance of protobuf-runtime is public registry, signature verified. Approved 2024-04-08 is the approval entry for protobuf-runtime. Linking mode matters for protobuf-runtime, which is used message encoding across edge and

cloud. protobuf-runtime was checked against the company's own register, which had not been updated since the change landed.

Safety-controller-toolchain shows an approval status of approved 2023-02-10 against the company's own policy. Safety-controller-toolchain carries a practical obligation: nothing beyond retaining the licence text with the source. Safety-controller-toolchain affects Safety Controller; no other component reviewed depends on it. safety-controller-toolchain was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Linking mode matters for pager-provider-sdk, which is used alarm dispatch. Pager-provider-sdk version 3.4.0 is catalogued at OSS-027 and is used alarm dispatch. Compliance with the pager-provider-sdk terms requires to keep the dynamic-linking boundary intact. pager-provider-sdk is the clearest instance in TDR-2026-25 of engineering practice being better than the documentation suggests.

Cell-model-lib was assessed for support as well as licence risk, and it carries a commercial supplier whose support statement expires inside the plan period. Tracing cell-model-lib to its source gives public registry, signature verified. Battery Management Service is where cell-model-lib is consumed, which determines whether distribution occurs at all. cell-model-lib was raised again with the owning engineer after the initial interview, and the position did not change.

On wheel-slip-model the reviewer records that a network-copyleft component in a prototype; the reviewer flagged the shipping risk before validation completes. Where wheel-slip-model came from matters here: public registry. Wheel-slip-model affects Wheel Odometry Filter; no other component reviewed depends on it. wheel-slip-model rests on a single source, and the reviewer has said so rather than implying corroboration.

One point about kafka-client is not resolved: one service still vendors a 6.x client. Meeting the terms on kafka-client means to publish the modified files it distributes. Version 3.6.1 of kafka-client is the catalogued build at OSS-015, used event bus producer and consumer clients. kafka-client sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

What the company must actually do about pointcloud-utils is to keep the dynamic-linking boundary intact. Beyond licensing, pointcloud-utils carries an upstream project with no active maintainer. Weak copyleft applies to pointcloud-utils, whose licence is MPL-2.0. pointcloud-utils was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

10. Security findings that constrain the capacity plan

What OTA Updater finding SEC-2025-09 enables is this: a single key holder could sign an unreviewed firmware artefact. For SEC-2025-09 against OTA Updater the record shows that key custody was evidenced by interview and the key ceremony record, which is strong evidence and is graded separately from severity. Residual risk after remediation of SEC-2025-09 against OTA Updater is medium; a two-person rule is designed but not enforced in the tool. On 2025-09-24 the tester recorded Medium-severity finding SEC-2025-09: oTA artefact signing keys held by three engineers without a quorum control. finding SEC-2025-09 would move from a concern to a verified defect if one further measurement were supplied.

Impact of Low-severity finding SEC-2026-18 is not theoretical; a diagnostic flood can slow pack telemetry reporting. Assessment finding SEC-2026-18 reduces to a residual risk of low; no fleet impact demonstrated. Strong evidence supports finding SEC-2026-18, namely that flood produced on a bench pack. Rating for assessment finding SEC-2026-18 is Low, band 3.4; the reviewer re-read the scoring rationale and accepted it. Battery Management Service finding SEC-2026-18 was checked against the company's own register, which had not been updated since the change landed.

Service tokens are issued with a twelve-hour lifetime is the substance of assessment finding SEC-2025-01, raised on 2025-09-16. SEC-2025-01 against Identity and Authorization produces this judgement: the remediation is adequate and the time to remediate is not. The remediation ledger shows assessment finding SEC-2025-01 as remediated, 2025-11-14. What remains after Identity and Authorization finding SEC-2025-01 is remediated is low once lifetimes are cut to one hour with refresh. High-severity finding SEC-2025-01 is included because a reader of TDR-2026-25 should be able to disagree with the inference without repeating the work.

Low is the rating Billing Usage Meter finding SEC-2026-10 carries, at 4.0 on the assessor's own scale. Low-severity finding SEC-2026-10 leaves residual risk that the reviewer assesses as low; object lock is applied and verified nightly. On assessment finding SEC-2026-10 the tester's evidence was that bucket configuration read from the cloud export, and the reviewer grades it limited, resting on a configuration read rather than on a demonstrated exploitation. Reading assessment finding SEC-2026-10, the defect is that billing usage ledger export written to a bucket without object lock; the discovery date is 2026-04-19. Low-severity finding SEC-2026-10 was one of two items where TDR-2026-25 could not obtain the underlying data within the engagement.

What remains after SEC-2025-12 is remediated is medium; management accepted the risk with a compensating audit trail. Assessment finding SEC-2025-12 records that analytics warehouse service account held write access to raw telemetry, discovered on 2025-09-25. Weighing assessment finding SEC-2025-12 against the compensating controls, the reviewer would not treat this as a closing condition and would put it on the first post-close sprint. The reason Low-severity finding SEC-2025-12 was not downgraded is that a loader defect or compromise could alter source telemetry. assessment finding SEC-2025-12 carries a follow-up request that was satisfied after the draft of TDR-2026-25 was circulated.

2025-11-27 is the closure position on assessment finding SEC-2025-04, which the ledger records as remediated. On assessment finding SEC-2025-04 the tester's evidence was that reproduced, and independently observed in a field incident, and the reviewer grades it moderate, because the demonstration was performed in a laboratory tenant rather than in production. What SEC-2025-04 enables is this: an authenticated author can distribute unreviewed geometry. SEC-2025-04 against Map Authoring Studio is rated Medium with a numeric band of 5.4 on the tester's scoring scale. finding SEC-2025-04 was one of two items where TDR-2026-25 could not obtain the underlying data within the engagement.

The residual position on assessment finding SEC-2025-08 is low; the interface is restricted to the operations subnet, stated after mitigation rather than assuming full success. On Medium-severity finding SEC-2025-08 the conclusion is that the item is open past its committed date and no revised date has been agreed. The evidence for SEC-2025-08 against Event Bus is moderate, because the demonstration was performed in a laboratory tenant rather than in production: reachability proven by the tester from a deployed container. On 2025-09-23 the tester recorded SEC-2025-08: event bus administrative interface reachable from the application subnet. SEC-2025-08 against Event Bus was reviewed against the position twelve months earlier, and the direction of travel is favourable.

2025-09-19 is the closure position on SEC-2025-02 against Release Orchestrator, which the ledger records as remediated. At 8.1 the score on SEC-2025-02 against Release Orchestrator places it in the High group. SEC-2025-02 matters because any engineer with build log access could replay a service credential. Release Orchestrator finding SEC-2025-02 is not fully retired by its remediation: low; redaction is enforced in the logging library. SEC-2025-02 against Release Orchestrator was inspected in the repository rather than described from the ticket, which is why the account is specific.

In High-severity finding SEC-2026-17 the exploitation path means a local compromise could alter planner limits without detection. Finding SEC-2026-17 stands remediated; 2026-06-19 is what the record gives for closure. The reviewer declines to soften High-severity finding SEC-2026-17: the fix is verified and the class of defect is not, since no test now prevents its return. The evidence for finding SEC-2026-17 is moderate: override applied on a laboratory robot and observed in plan output. High-severity finding SEC-2026-17 illustrates why TDR-2026-25 separates the severity of a finding from the confidence behind it.

SEC-2025-03 against Remote Operations Console rests on the observation that demonstrated in the tester's laboratory network, not on a customer site; the reviewer holds that to be strong. SEC-2025-03 is remediated, with a closure date of 2026-01-22. Impact of SEC-2025-03 against Remote Operations Console is not theoretical; an on-path attacker on a customer network could observe operator commands. Rating for SEC-2025-03 against Remote Operations Console is Medium, band 6.2; the reviewer re-read the scoring rationale and accepted it. assessment finding SEC-2025-03 was excluded from an earlier draft of TDR-2026-25 and reinstated after the reviewer obtained the supporting record.

Assessment finding SEC-2026-13 records that order broker accepted an oversized payload without a limit, discovered on 2026-04-20. Finding SEC-2026-13 matters because a large payload degrades broker throughput. Residual exposure on SEC-2026-13 reads low. Low-severity finding SEC-2026-13 arrives as a Low finding with a 3.3 score, and

the reviewer did not adjust either. finding SEC-2026-13 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Low is the rating SEC-2026-11 carries, at 4.4 on the assessor's own scale. Gripper Controller finding SEC-2026-11 leaves residual risk that the reviewer assesses as medium; the safety controller limit is the compensating control. The finding at Low-severity finding SEC-2026-11, dated 2026-04-19, is that gripper controller parameter store writable by any on-robot process. For SEC-2026-11 the reviewer notes that the risk acceptance lacks an expiry date, which is a process finding in its own right. assessment finding SEC-2026-11 was checked against the company's own register, which had not been updated since the change landed.

Identity and Authorization finding SEC-2026-08 is supported by the fact that role grant performed in a staging tenant; evidence strength is strong. What SEC-2026-08 against Identity and Authorization enables is this: one administrator can grant themselves tenant-wide access. After the fix, Identity and Authorization finding SEC-2026-08 carries residual risk described as low; role grants now require a second approver and are alerted. Severity on Identity and Authorization finding SEC-2026-08 is Medium, scored 6.4 by the assessor. Medium-severity finding SEC-2026-08 is the entry that changed most between the first and second document requests behind TDR-2026-25.

11. Capacity qualifications and unresolved questions

The open items carried forward from this report are set out below; a reader should treat the committed dates as intentions rather than facts. The open items in report TDR-2026-25 are grouped by what would resolve them rather than by which team owns them, because several cross team boundaries. One declinature in this report rests on an engineering judgement the reviewer shares; the others rest on prioritisation, which is a different kind of argument. Not every finding in this report is accepted by management: 4 are declined, and the declinatures are reproduced rather than argued away. Calder Rowe Engineering Advisory has not been asked to opine on valuation, and nothing in the report at TDR-2026-25 should be read as doing so.

The report at TDR-2026-25 distinguishes three positions: verified defect, credible concern, and matter requiring further work. The third category is not empty. Grading in the present report is deliberately coarse, because a finer scale would imply a precision the underlying evidence does not carry. M. Quillan led the work behind the report at TDR-2026-25, combining document review, 22 interviews and hands-on inspection of running systems. Evidence gathering for TDR-2026-25 was iterative: an initial document request, 22 interviews, then a second request that closed most of the gaps. Calder Rowe Engineering Advisory has flagged 2 matters for the transaction record and left the remainder as engineering work.

The present report keeps declined items visible: risk acceptance without an expiry date is itself recorded as a finding. 4 items in TDR-2026-25 sit in a declined state, and the reviewer regards one of them as wrongly declined. M. Quillan could not verify one class of claim in this report, namely the behaviour of components deployed only at the two sites not visited. Report TDR-2026-25 carries one structural limitation: the company's own metrics are the only source for several objectives, and the reviewer could not build an independent measurement inside the engagement. Any figure in the report at TDR-2026-25 that the reviewer could not trace to a system is attributed to the person who gave it.

This report would have the monitoring pack include declined items with their compensating controls, so that acceptance decisions age visibly. M. Quillan notes in report TDR-2026-25 that the company's own reporting is adequate for three of the four recommended indicators. Nothing in this report is presented as verified on the strength of an interview alone. This report grades evidence as strong, moderate or limited, and a limited grade is not upgraded because the consequence would be serious. Readers comparing this report with an earlier technical report should note that the service inventory was renamed in between.

12. Conclusions on capacity against the operating plan

Whatever weight a reader gives the individual findings, report TDR-2026-25 concludes that the engineering claims that matter to the investment case are supportable, and two of the claims the company leads with are not. Taking the sections above together, the engineering claims that matter to the investment case are supportable, and two of the claims the company leads with are not. Report TDR-2026-25 recommends that the remediation commitment register be reported to the board quarterly with evidence attached, not just status. Calder Rowe Engineering Advisory recommends that the 6

indicators be reported with their measurement basis attached, since two of them changed basis during the reviewed period. The report at TDR-2026-25 does not restate the company's own reporting where the reviewer was able to verify the underlying data instead.

M. Quillan would put 8 items from the report at TDR-2026-25 into the first hundred days rather than into a warranty. TDR-2026-25 suggests one governance change rather than a price adjustment: a quarterly technical review with the lead investor's observer present. The report at TDR-2026-25 follows the reviewer's usual practice of grading evidence strength separately from finding severity, so a plausible concern is not promoted to a verified defect. The approach M. Quillan took in the present report was to test the company's own assertions rather than to restate them. This report was prepared on the footing that a reader may be an engineer or may not, and defines terms where the distinction matters.

Reporting under report TDR-2026-25 should distinguish a target that was met from a target that was lowered, which the current pack does not. 6 indicators are recommended in TDR-2026-25; the reviewer deliberately avoided proposing anything that needs a new pipeline. M. Quillan has dated each open item in report TDR-2026-25 so that a reader can see which have been open longest rather than only how many there are. This report preserves 5 unresolved questions rather than resolving them by assumption. TDR-2026-25 is one input to a diligence process and is not a substitute for the company's own engineering judgement.